

Day 11 – Hands-on Lab Practice

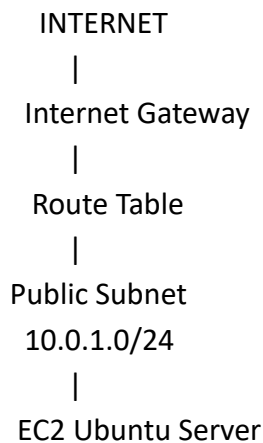
EC2 Creation with Expected Errors & Troubleshooting Notes

Lab Objective

Students will:

- Launch EC2 instances
 - Configure networking
 - Attach Security Groups
 - Connect using SSH
 - Install applications
 - Troubleshoot common EC2 issues
-

Lab Architecture



Lab 1 – Create EC2 Instance

Step 1 – Open EC2 Dashboard

Go to:

AWS Console → EC2 → Launch Instance

Step 2 – Configure Instance

Option	Value
Name	dev-server
AMI	Ubuntu 24.04
Instance Type	t2.micro
Key Pair	create/select existing
VPC	custom-vpc
Subnet	public-subnet
Public IP	Enable

Expected Error 1 – No Key Pair Selected

Error

Cannot launch instance without key pair

Reason

SSH authentication requires .pem key.

Solution

Create new key pair:

EC2 → Key Pairs → Create Key Pair

Download:

mykey.pem

Step 3 – Configure Security Group

Inbound Rules:

Type	Port	Source
SSH	22	My IP
HTTP	80	Anywhere
HTTPS	443	Anywhere
Custom TCP	8080	Anywhere

Expected Error 2 – SSH Connection Timeout

Error

ssh: connect to host timeout

Possible Reasons

Reason	Description
Port 22 blocked	Security group issue
No Public IP	Instance inaccessible
Route table missing	No internet path
IGW not attached	Internet unavailable

Troubleshooting Steps

Verify Security Group

Check inbound rule:

SSH → Port 22 → My IP

Verify Public IP

Go to:

EC2 → Instance → Networking

Check:

Public IPv4 Address

Verify Route Table

Must contain:

Destination Target

0.0.0.0/0 IGW

Verify IGW Attachment

Go to:

VPC → Internet Gateway

Status should be:

Attached

Step 4 – Connect to EC2

Linux/Mac:

```
chmod 400 mykey.pem
```

```
ssh -i mykey.pem ubuntu@PUBLIC-IP
```

Windows:

Use:

- PuTTY
 - MobaXterm
 - Git Bash
-

Expected Error 3 – Permission Denied (Publickey)

Error

Permission denied (publickey)

Reasons

Issue	Fix
Wrong username	Use ubuntu/ec2-user
Wrong PEM file	Select correct key
Key permission issue <code>chmod 400</code>	

Correct Usernames

AMI	Username
Ubuntu	ubuntu
Amazon Linux	ec2-user
CentOS	centos

Step 5 – Install NGINX

```
sudo apt update
sudo apt install nginx -y
```

Start service:

```
sudo systemctl start nginx
sudo systemctl enable nginx
```

Expected Error 4 – Unable to Access Website

Error

Browser shows:

This site can't be reached

Troubleshooting

Check HTTP Rule

Security Group should allow:

Type Port

HTTP 80

Verify NGINX Status

```
sudo systemctl status nginx
```

Expected:

active (running)

Verify Port Listening

```
sudo netstat -tulnp
```

OR

```
sudo ss -tulnp
```

Check:

0.0.0.0:80

Step 6 – Modify Custom Web Page

```
cd /var/www/html
```

```
sudo vi index.html
```

Example:

```
<h1>Welcome to AWS DevOps Lab</h1>
```

Access:

http://PUBLIC-IP

Expected Error 5 – EC2 Has No Internet

Symptoms

ping google.com

Fails.

Reasons

Problem	Description
No IGW	Internet unavailable
Wrong route table	Traffic blocked
Private subnet	No public access
NACL issue	Network blocked

Fix

Ensure:

- IGW attached
 - Route table configured
 - Public subnet association done
 - Public IP enabled
-

Step 7 – Install Apache Tomcat

```
sudo apt install default-jdk -y
```

Download Tomcat:

```
wget https://downloads.apache.org/tomcat/tomcat-9/v9.0.105/bin/apache-tomcat-9.0.105.tar.gz
```

Extract:

```
tar -xvzf apache-tomcat-9.0.105.tar.gz
```

Start:

```
cd apache-tomcat-9.0.105/bin
sh startup.sh
```

Access:

http://PUBLIC-IP:8080

Expected Error 6 – Port 8080 Not Accessible

Reason

Security group missing custom TCP rule.

Fix

Add inbound rule:

Type	Port
------	------

Custom TCP 8080	
-----------------	--

Source:

0.0.0.0/0

Step 8 – Attach Elastic IP

Go to:

EC2 → Elastic IPs

Allocate:

New Elastic IP

Associate with EC2.

Why Elastic IP?

Without Elastic IP:

- Public IP changes after stop/start.

With Elastic IP:

- Fixed public IP.
-

Expected Error 7 – Website Down After Restart

Reason

Public IP changed.

Solution

Use:

- Elastic IP
 - Route53
 - Load Balancer
-

Real-Time Production Notes

Service	Recommended Setup
Web Server	Public Subnet
Database	Private Subnet
Application Server	Private/Public based on design
Bastion Host	Public Subnet
Load Balancer	Public Subnet

Important Commands Cheat Sheet

Check IP

ip a

Check Disk

df -h

Check Memory

free -m

Check Running Services

systemctl list-units --type=service

Check Open Ports

ss -tulnp

Day 12 Hands-on Tasks

Task 1

Create EC2 in custom VPC.

Task 2

Install NGINX.

Task 3

Install Tomcat on port 8080.

Task 4

Attach Elastic IP.

Task 5

Troubleshoot:

- SSH issue
- HTTP issue
- Port 8080 issue